Tutorial 02 — Instructor notes

ENVX2001 – Applied Statistical Methods

Semester 1

Contents

Companion document for tutors running Tutorial 02. Mirrors each exercise with talking points, expected output at each checkpoint, and common problems.

Student document: Tutorial 02

Before you start

Get these sorted before students arrive:

- Have RStudio open with a fresh session.
- Make sure the `DAAG` package is installed. Run `install.packages("DAAG")` if you have not done so already — you do not want to troubleshoot installations during the tutorial.
- Load the possum dataset and check it renders: `library(DAAG); data(possum)`. You will refer to it throughout.
- Skim the student document if you have not recently. The two exercises build on each other, so you will need to keep the thread going.

Exercise 1: What confidence intervals tell us

Part A — Compute and interpret

What to do

Load the dataset and run through `str(possum)` to show the structure. Then compute the 95% CI for head length using `t.test(possum$hdlngth)`.

```
CODE
library(DAAG)
data(possum)
str(possum)
```

```
CODE
t.test(possum$hdlngth)
```

What to say

- Walk through the `t.test()` output: sample mean (92.6 mm), confidence interval [91.9, 93.3], and the t-statistic. Students do not need to worry about the t-statistic yet — focus on the mean and the interval.
- Spend time on the interpretation. The most common mistake is saying "there is a 95% probability the true mean is in this interval." That is not what it means. The true mean is fixed — it is either in the interval or it is not.
- The correct interpretation: if we repeated this study many times, about 95% of the intervals we compute would contain the true mean. The 95% refers to the method's long-run success rate.

- Use an analogy if it helps: a net that catches the fish 95% of the time. Once you have cast the net, the fish is either in it or not — but you trust the net because it works most of the time.

Expected output

`t.test()` prints the sample mean (92.6 mm), the 95% CI [91.9, 93.3], and a t-statistic with p-value (which students can ignore for now).

Common issues

> ⚠ Warning
>
> **"could not find function 'data'"** Student has not loaded the `DAAG` package. They need `library(DAAG)` before `data(possum)`.

Part B — What affects the width?

What to do

Draw random samples of 10 and 50 possums and compute confidence intervals for each. Compare widths across n = 10, n = 50, and the full n = 104.

CODE

```
set.seed(2026)
samp10 <- sample(possum$hdlngth, 10)
t.test(samp10)
```

CODE

```
set.seed(2026)
samp50 <- sample(possum$hdlngth, 50)
t.test(samp50)
```

What to say

- The key takeaway is that more data gives a narrower interval. With 10 possums the interval is roughly 5.6 mm wide. With 50 it is about 2.0 mm. With all 104 it is 1.4 mm.
- Explain why: the standard error is SD / $\sqrt{n}$. As n increases, the standard error shrinks, and the interval gets tighter.
- `set.seed()` makes the random sample reproducible. If students use a different seed, they will get a different sample — that is fine, but their numbers will not match yours exactly.
- If time allows, invite a few students to change the seed and share their interval widths. This reinforces the idea that every sample gives a slightly different interval.

Expected output

Three confidence intervals with decreasing widths: ~5.6 mm (n = 10), ~2.0 mm (n = 50), ~1.4 mm (n = 104).

Common issues

> ⚠ Warning
>
> **Different numbers from the student document.** Students who forget `set.seed(2026)` or use a different seed will get different sample means and interval widths. The pattern (wider with fewer observations) still holds — reassure them that the exact numbers do not matter.

> ⚠ Warning
>
> **"Why not always use the full dataset?"** Good question. In practice we rarely have the entire population — we work with whatever sample we can collect. The point of this exercise is to show that sample size affects precision, which matters when you are designing a study.

Exercise 2: When groups matter

Part A — The pooled estimate

What to do

Compute a confidence interval for ear conch length across all 104 possums.

CODE

```
t.test(possum$earconch)
```

What to say

- The mean ear conch length across all possums is about 48.1 mm, with a 95% CI of [47.3, 48.9].
- This looks like a reasonable estimate — until we check whether the data has structure we are ignoring.
- Mention that these possums come from two distinct populations: Victorian sites and sites further north (NSW/QLD). Set up the next part by asking: "What happens if the two groups have different ear sizes?"

Expected output

`t.test()` output showing mean ≈ 48.1 mm, 95% CI ≈ [47.3, 48.9].

Part B — Split by population

What to do

Compute separate confidence intervals for Victorian and non-Victorian possums, then show the boxplot.

```
CODE
t.test(possum$earconch[possum$Pop == "Vic"])
```

```
CODE
t.test(possum$earconch[possum$Pop == "other"])
```

```
CODE
library(ggplot2)

ggplot(possum, aes(x = Pop, y = earconch, fill = Pop)) +
  geom_boxplot(show.legend = FALSE) +
  labs(x = "Population", y = "Ear conch length (mm)") +
  theme_bw()
```

What to say

- Victorian possums have a mean ear conch length of about 52.2 mm. The other population averages about 44.9 mm — over 7 mm shorter.
- The pooled CI of [47.3, 48.9] sits between the two groups. It does not describe either population accurately. It is an average of two different things.
- The boxplot makes this obvious visually. Point out that the boxes barely overlap — these are genuinely different populations.
- This is a classic example of where ignoring group structure gives a misleading summary. The overall mean is not "wrong" — it is just not useful if you care about either group specifically.

Expected output

Two separate CIs that do not overlap (Vic ≈ [51.5, 52.9], other ≈ [44.5, 45.3]). A boxplot showing clear separation between the two populations.

Common issues

> ⚠ Warning
>
> **Subsetting syntax confusion.** `possum$earconch[possum$Pop == "Vic"]` is dense for students seeing it for the first time. Break it down: `possum$Pop == "Vic"` creates a logical vector, and the square brackets select the matching rows. If students struggle, show the logical vector first.

> ⚠ Warning
>
> **ggplot2 not loaded.** Students who skipped the `library(ggplot2)` line will get "could not find function 'ggplot'". The student document includes the library call in the code block, but some students strip it out when typing.

Part C — Connection to the lab

What to do

This is a short wrap-up linking to the lab. No code to run — just a brief discussion.

What to say

- In the lab, students will learn stratified random sampling, which handles exactly this kind of structure.
- The idea is straightforward: estimate each subgroup separately, then combine the estimates in a way that accounts for the group sizes.
- This gives a more accurate overall estimate than pooling everything together and pretending the groups do not exist.
- You do not need to teach stratification here — just plant the seed so students recognise the problem when they see the lab exercise.

Putting it all together

What to say

- The two big ideas from today: (1) confidence intervals get narrower with more data, and (2) ignoring group structure can give misleading estimates.

- Both of these come up in the lab this week. The CI width idea connects to sample size planning. The group structure idea connects to stratified sampling.
- If students ask “when should I split by group?” — a good rule of thumb is to check whether there is a known grouping variable in the data. If the groups look different (as the boxplot showed), a pooled estimate can be misleading.